First Progress Review

Chapter 1: Introduction

Calibrated Selective Autonomy in Multi-Agent Clinical Artificial Intelligence: A Clinician-Validated, Privacy-Preserving Framework for Brain-Tumour Diagnosis and Surgical-Planning Support in Low-Resource Settings

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Draft version 1 · 10 November 2026

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1. Introduction

This chapter introduces the research for the First Progress Review. It restates the background and the problem that motivate the study, updates them with evidence gathered during the first months of candidature, and sets out the aim, objectives, research questions, scope and significance of the work. The chapter is based on the research proposal accepted by the Board on 4 November 2026 and on the literature collected in the Research Paper Bank (559 open-access papers grouped by objective; see Chapter 2). Where the evidence reviewed since registration sharpens or qualifies a claim made in the proposal, this is stated explicitly.

1.1 Background of the Research

Artificial intelligence (AI) now supports diagnosis, treatment planning and clinical documentation at a scale no human reader could match (Topol, 2019; Khalighi et al., 2024). In medical imaging alone, a meta-analysis of 503 studies found high diagnostic accuracy for deep learning across ophthalmology, breast and respiratory imaging, while noting that very few of those studies were prospective or externally validated (Aggarwal et al., 2021). The same capability makes unsupervised deployment hazardous. Performance degrades under dataset shift: a pneumonia detector generalised poorly outside its training hospitals because it had learned site-specific artefacts rather than disease (Zech et al., 2018; Finlayson et al., 2021). And a model that reports a confident answer gives the

clinician no signal that it has left the region in which it is competent (Kompa et al., 2021; Guo et al., 2017).

These findings have made human-in-the-loop (HITL) AI the dominant design philosophy for safety-critical medicine. HITL inserts control points at consequential moments so that a clinician can validate, refine or override a recommendation using findings, comorbidity and patient preferences that no model can see (Olawade et al., 2026). The evidence reviewed for this progress report shows, however, that simply placing a human beside an AI is not enough. A meta-analysis of 106 human-AI studies found that, on average, human-AI combinations performed worse than the better of the human or the AI alone in decision tasks (Malone et al., 2024); radiologists' benefit from AI assistance varied widely and incorrect AI output harmed their performance (Yu et al., 2024); and explanations increased acceptance of wrong recommendations (Bansal et al., 2021). By contrast, systems that decide *which* cases the AI should handle and which should go to the clinician have improved both accuracy and workload: complementarity-driven deferral in mammography and tuberculosis screening reduced false positives and clinician reads at the same time (Dvijotham et al., 2023). The open engineering question, and the one this research addresses, is therefore not whether a human should be in the loop, but when, on which cases, and on what calibrated evidence.

1.1.1 The brain-tumour pathway and the state of computational support

The brain-tumour pathway is long, information-dense and unforgiving. It runs from non-specific neurological symptoms and a scan, through multi-sequence magnetic resonance imaging (MRI), biopsy and microscopic examination of tissue, laboratory tests on that tissue, a multidisciplinary decision, surgery and follow-on therapy, to years of surveillance scanning (Figure 1.1). Since 2021 the World Health Organization (WHO) classification has been integrated, so a diagnosis combines tissue appearance with molecular results (Louis et al., 2021), making the task multi-source by definition. Brain tumours carry disproportionate mortality and affect working-age adults and children (Price et al., 2024). Surgery must balance removing as much tumour as possible against preserving the regions controlling speech, movement and vision: extent of resection is repeatedly associated with survival (Sanai & Berger, 2008; Molinaro et al., 2020), while removal of healthy tissue causes permanent loss of function. The pathway is also dependent: an error in outlining the tumour propagates into grading, grading into treatment selection, and treatment selection into outcome.

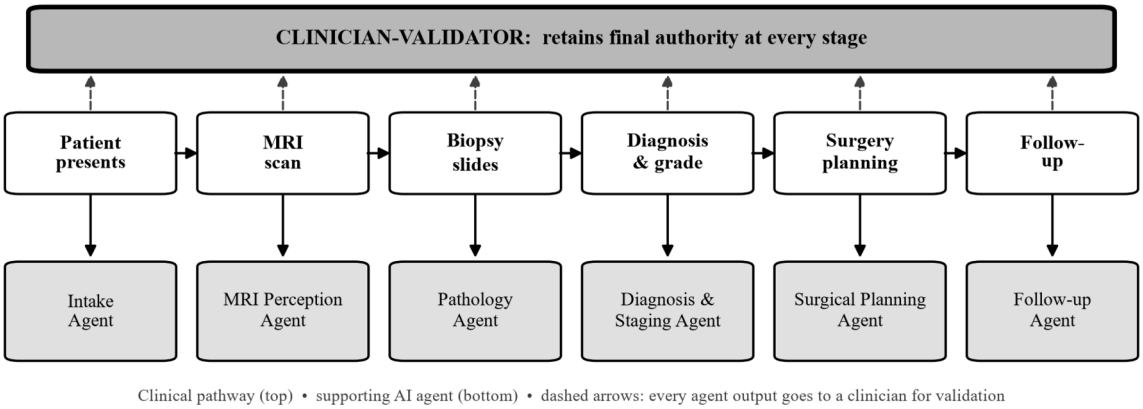


Figure 1.1: The brain-tumour care pathway, the specialised AI agent that supports each stage, and the clinician who retains final authority throughout (from the research proposal).

Computational support has advanced rapidly for individual sub-tasks. Deep learning dominates tumour detection, outlining and classification on MRI, benchmarked on the BraTS challenges (Menze et al., 2015; Baid et al., 2021), with nnU-Net establishing strong, self-configuring baselines (Isensee et al., 2021) that remain hard to beat when validated rigorously (Isensee et al., 2024). Computational pathology reaches clinical-grade analysis of whole digitised slides with weak, slide-level labels (Lu et al., 2021), and deep learning has achieved neuropathologist-level integrated classification of adult-type diffuse gliomas (Wang et al., 2023). Fusion of pathology and radiology foundation models now supports WHO 2021 glioma subtyping (Saueressig et al., 2026), and molecular classification has been brought into the operating theatre (Hollon et al., 2023; Vermeulen et al., 2023). Yet individual models solve individual tasks: no accepted architecture assembles them into a supervised workflow with defined accountability.

Two further developments frame this work. The first is agentic and multi-agent AI, in which a coordinating component routes work between specialised models or language-model agents. Multi-agent systems have outperformed single models on medical benchmarks (Kim et al., 2024; Chen et al., 2025), and an autonomous agent working inside a simulated electronic health record recently exceeded physicians on emergency cases (Ferber et al., 2026). However, a scoping review found only seven eligible studies of agentic AI in healthcare and a single randomised trial (Collaco et al., 2026), and audits of multi-agent systems show high failure rates and false consensus (Cemri et al., 2025; Zhu et al., 2025). The second is federated learning, which lets institutions train a shared model without disclosing raw data and has been proven for glioblastoma boundary detection across 71 sites on six continents (Pati et al., 2022).

1.1.2 The Sri Lankan context and the case for a national registry

GLOBOCAN 2024 estimates approximately 700 new brain and nervous-system cancer cases and around 587 deaths annually in Sri Lanka, with a five-year prevalence of roughly 2,500 (International Agency for Research on Cancer, 2024). Deaths are high relative to new cases, and care is concentrated in a few tertiary units served by a small pool of neurosurgeons, neuroradiologists and neuropathologists. A National Cancer Registry exists for incidence and mortality reporting (National Cancer Control Programme, 2024), but there is no imaging- and tissue-linked brain-tumour registry; patients are tracked in fragmented, institution-specific records. Clinically this impedes continuity of care, multidisciplinary review and outcome auditing. Scientifically it makes locally valid AI almost impossible to build, because models trained on foreign cohorts do not transfer reliably to local scanners, protocols, laboratory practice or case mix.

Evidence from comparable settings supports this concern. The BraTS-Africa challenge was created because routine-care MRI in Sub-Saharan Africa, often from older or low-field scanners with poorer contrast and from patients presenting later, challenges models developed on Western benchmark data; the winning teams needed heavy augmentation and ensembling to cope with a small local dataset (Adewole et al., 2026; Ankomah et al., 2025), and reviews of AI in low- and middle-income countries report few real-world evaluations, limited data and poor fit to local workflows (Ciecierski-Holmes et al., 2022; Yevudza et al., 2024). The project therefore separates two layers. The **platform layer** is Sri Lanka's first National Brain Tumour Patient Registry, which gives the research authentic data and users and is a real-world deliverable in its own right. The **research layer**, the focus of this PhD, is the design, formalisation and validation of the AI framework that makes such a platform safe, data-efficient and privacy-preserving.

1.2 Problem Statement and Research Gap

1.2.1 Why accuracy alone is not enough

A model that is 95% accurate is not automatically safe: the danger lies in the 5% of confident mistakes, and in the absence of a mechanism by which the system can recognise the limits of its own competence. Modern deep networks are systematically over-confident unless explicitly calibrated (Guo et al., 2017), and calibration worsens precisely when a model meets data from a new hospital (Ovadia et al., 2019). Benchmarks of failure detection in medical imaging found that no published confidence-scoring method reliably

prevented silent failures (Bungert et al., 2023; Bernhardt et al., 2022). The issue is therefore not a shortage of accurate models but the absence of a governing mechanism that decides, case by case, whether an output is fit to be acted upon. Where specialist review time is the scarcest resource, that mechanism also determines whether AI reduces clinician workload or merely relocates it (Chen et al., 2024).

1.2.2 Identified research gaps

The proposal identified six gaps. The literature reviewed since registration (Chapter 2) confirms each of them and adds specific evidence, summarised in Table 1.1.

Table 1.1: Research gaps, the evidence that confirms them, and the objective that addresses each.

Gap	What the literature shows	Objective
No principled policy for when to defer	Selective classification, learning to defer and conformal prediction exist but address a single model and a single decision with a uniform error cost (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Angelopoulos & Bates, 2021); cost- and capacity-aware deferral has been studied only outside medicine or on simulated experts (Alves et al., 2024; Verma & Nalisnick, 2022).	O2
Unqualified hand-offs between agents	Medical multi-agent systems route by question difficulty or LLM self-rating, not calibrated uncertainty (Kim et al., 2024; Lou et al., 2026); unsupported observations propagate downstream in 16.6% of cases (Zhu et al., 2025); optimising components does not optimise the system (Bedi et al., 2025).	O1
Fusion that does not survive real data	Fusion beats single sources on complete cohorts (Chen et al., 2022; Saueressig et al., 2026) but most methods assume every source is present; missing-modality work is simulated on BraTS (Zhang et al., 2022; Liu et al., 2023).	O3
Methods that assume abundant, standardised data	Public benchmarks are mostly Western and pre-processed (Baid et al., 2021); routine-care MRI from low-resource settings needed a dedicated benchmark and adaptation (Adewole et al., 2026; Ankomah et al., 2025); transfer and self-supervised learning help with limited data (Boyd et al., 2024; Tang et al., 2022).	O3
Data sharing in a few small, heterogeneous hospitals	Federation works at international scale (Pati et al., 2022; Zenk et al., 2025) but the effect of aggregation on calibration, and the small-federation regime, are little studied (Manthe et al., 2024; Koutsoubis et al., 2024).	O3
No way to measure the quality of the escalation decision	Reporting guidelines require human factors and safety in situ (Vasey et al., 2022; Sounderajah et al., 2025) but do not define escalation-quality metrics; selective prediction can change the pattern of clinician errors (Jabbour et al., 2025).	O4

1.2.3 Problem statement

These gaps identify the central research problem as the absence of a clinically responsible, formally governed multi-agent AI framework that can support the brain-tumour pathway across imaging, tissue analysis, combined diagnosis and pre-operative surgical planning while explicitly managing uncertainty and preserving clinician authority. Such a framework must decide, for each case and each sub-decision, when an agent can provide

reliable decision support and when the case must be escalated to a qualified clinician. It must remain robust to the small, heterogeneous, non-standardised datasets of Sri Lankan institutions, and enable privacy-preserving collaboration without centralising raw patient data.

1.3 Aim, Objectives and Research Questions

1.3.1 Aim

The aim is to design, develop and evaluate an uncertainty-aware, human-in-the-loop multi-agent AI framework for brain-tumour care, integrating imaging, tissue analysis, combined clinical decision support and surgical planning, and to evaluate its feasibility within a national brain-tumour registry in Sri Lanka. Human oversight is implemented as an explicit, measurable mechanism rather than treated as a guiding principle.

1.3.2 Research objectives

The objectives are unchanged from the accepted proposal and are deliberately limited to four:

- **O1.** To design and develop a multi-agent AI architecture for the brain-tumour care pathway (patient intake, MRI analysis, tissue analysis, diagnosis and staging, surgical planning and follow-up), in which an orchestrator routes tasks, propagates uncertainty across agent hand-offs, and keeps an auditable record of every AI output and every clinician validation.
- **O2.** To develop and evaluate a calibrated, uncertainty-aware escalation mechanism that decides when an agent may provide a decision-support output and when a case or sub-decision must be referred to a clinician, based on selective prediction, learning to defer and the clinical cost of error.
- **O3.** To develop and evaluate the data-efficient models the agents depend on (MRI detection, outlining, classification and anatomical reconstruction; analysis of digitised biopsy slides; and their combination with available laboratory markers) on small, heterogeneous Sri Lankan data, including privacy-preserving federated training across institutions, and to benchmark their performance and calibration against established approaches.
- **O4.** To clinically evaluate the human-in-the-loop framework, covering diagnostic and decision-support performance, the appropriateness of escalation decisions, clinician

workload, usability, trust and safety, with surgical planning evaluated as a proof-of-concept component.

1.3.3 Research questions

The research questions correspond one-to-one with the objectives:

- **RQ1.** How can specialised AI agents be orchestrated across the brain-tumour care pathway so that uncertainty is propagated between them, information is coordinated, and an auditable record of AI outputs and clinician validations is maintained?
- **RQ2.** How can uncertainty estimation and calibration be incorporated into the framework to determine when the system should provide a decision-support output and when it should defer to a clinician?
- **RQ3.** How effectively can data-efficient and multi-source methods detect, outline, classify and grade brain tumours from small and heterogeneous Sri Lankan MRI and biopsy-slide data, including under federated training across hospitals, and how do their performance and calibration compare with established approaches?
- **RQ4.** To what extent can the human-in-the-loop framework improve clinically relevant decision support while maintaining appropriate clinician involvement, acceptable workload, usability, trust and safety, including the feasibility of MRI-based surgical-planning support?

1.4 Scope, Limits of Autonomy and Limitations

1.4.1 Scope

Depth of investigation is differentiated across three tiers. The first and primary tier is the uncertainty-aware orchestration and escalation mechanism, supported by data-efficient MRI models; this is the central methodological contribution. The second is computational pathology and combined decision support, which remain decision support rather than autonomous diagnosis. The third is privacy-preserving federated learning. The surgical-planning component is a proof of concept covering patient-specific anatomical reconstruction, outlining of the tumour and structures at risk, and visualisation of surgical risk; it makes no autonomous surgical decisions. The registry is enabling infrastructure. The study is limited to participating Sri Lankan institutions and to tumours included in the registry under predefined eligibility criteria.

1.4.2 Limits of AI autonomy

Autonomy is conditional, uncertainty-aware and task-specific. The system does not independently diagnose, decide treatment or perform surgery. Each agent produces a decision-support output with a calibrated uncertainty; cases exceeding predefined thresholds are escalated, and the final decision remains the responsibility of the treating clinician. The system never performs or controls an operation: it plans, predicts, flags and recommends, and the neurosurgeon executes and validates every consequential step. This follows international guidance that places human autonomy, transparency and accountability among the core principles of AI for health (World Health Organization, 2021), the requirement in the EU AI Act that high-risk systems be designed so that people can oversee and interrupt them (European Union, 2024), and the FUTURE-AI consensus on traceability and usability (Lekadir et al., 2025). Three boundaries follow: bounded decision authority (no diagnosis, grade or treatment decision of record is issued by the AI); a conservative default (under uncertainty or novelty the system defers); and transparency of basis (every autonomous action carries its evidence, confidence and a model-facts statement of the conditions under which it was validated (Sendak et al., 2020; Mitchell et al., 2019)).

1.4.3 Limitations and assumptions

Component performance may vary across tumour types, imaging protocols, institutions and patient populations, and the federated component depends on the number of participating institutions, computing infrastructure and connectivity. Intra-operative guidance, robotic actuation, new laboratory assays, radiotherapy and chemotherapy planning, paediatric-specific modelling and medical-device certification are outside scope. Three further limitations are stated plainly. **Cohort size:** national incidence means the registry will accrue hundreds rather than thousands of fully linked cases, which bounds subgroup analyses and makes data-efficient methods necessary. **Reference standard:** clinician-confirmed diagnoses serve as ground truth, so the framework is evaluated against expert consensus. **Study type:** this is not a randomised trial of patient outcomes; validation measures diagnostic performance, workload, escalation appropriateness, safety and trust in an assistive workflow (Vasey et al., 2022), and no survival benefit is claimed. The work assumes that ethics approval and data-sharing agreements are secured before data access (the KDU application was submitted on 10 November 2026), that units can supply usable

MRI and digitisable biopsy material, and that clinician time is available, though scarce. Each assumption has a planned mitigation.

1.5 Justification and Significance

Scientific. How an AI agent should decide when it is competent to act and when it must defer is a computer-science problem before it is a clinical one. It sits at the intersection of uncertainty quantification (Gal & Ghahramani, 2016; Lakshminarayanan et al., 2017), calibration (Guo et al., 2017; Ovadia et al., 2019), selective prediction and learning to defer (Geifman & El-Yaniv, 2017; Mozannar & Sontag, 2020; Dvijotham et al., 2023) and multi-agent orchestration (Liu et al., 2025). The literature treats these as separate concerns. Formalising selective autonomy over a chain of dependent sub-decisions, and evaluating it on an explicit workload-versus-safety frontier, is methodologically novel and transferable beyond this clinical domain. Recent preprints that add conformal risk control to a single routing step (Li et al., 2026) or a safety governor to a tumour-board agent (Liu et al., 2026) show that the field is moving in this direction, which makes a rigorous, clinically validated treatment timely.

Clinical and national. Brain-tumour care is an unusually strong test case: it requires integration of several evidence sources (Louis et al., 2021), uncertainty must propagate along a sequential chain, the cost of error is severe and asymmetric (Sanai & Berger, 2008; Molinaro et al., 2020), and specialist review time is the binding constraint. Establishing Sri Lanka's first imaging- and tissue-linked brain-tumour registry delivers durable value to the participating units independently of the research findings, and an associated de-identified benchmark addresses the near-absence of South Asian cases in public brain-tumour datasets (Baid et al., 2021; Adewole et al., 2026).

Ethical, regulatory and methodological. A framework that makes oversight measurable and auditable rather than declared addresses a requirement that is becoming mandatory (European Union, 2024; Sounderajah et al., 2025). The constraints of the setting also make the results generalisable: small, heterogeneous data and barriers to centralisation are the normal condition of clinical AI in most of the world, so methods demonstrated under scarcity, heterogeneity and federation reach further than those requiring large curated corpora (Pati et al., 2022; Ciecierski-Holmes et al., 2022).

1.6 Progress Since Registration

The candidature started on 1 October 2026. In the first six weeks: the proposal was accepted by the Board and handed over to the evaluators (4 November 2026); the literature collection for all four objectives was completed and organised into the Research Paper Bank, with methodology extracted from each paper; the ethics application to the Research Ethics Committee, Faculty of Medicine, General Sir John Kotelawala Defence University (KDU), was finalised, signed by all investigators and submitted with the processing fee on 10 November 2026, to be followed by the University of Sri Jayewardenepura application; and a progress-tracking website was set up for the supervisors. These are reported in detail in Chapter 4.

1.7 Structure of the Report

Chapter 2 reviews the literature by objective and identifies the open problems. Chapter 3 sets out the research methodology: design-science research with a mixed-methods evaluation, the agent architecture and escalation rule, the work packages, data, analysis and ethics. Chapter 4 reports preliminary work and results.

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